Investment and chronological operation of electricity storage and electrolysis

Methodology, Modelling and Consulting Skills
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Abstract

This study considers investment in batteries and electrolysers for the constructed energy system supplied in the project. It combines a deterministic investment appraisal (Tier 1) with a forecast-based chronological extension (Tier 2B). Before modelling, the five course datasets are reconciled to a common daily calendar. Missing generation records, query-endpoint snapshots, invalid demand observations and price-aggregation errors are repaired using archived upstream records, calibrated alternative series and documented imputation. Research on equipment costs and hydrogen support values provides a common financial basis.

Tier 1 is formulated as a mixed-integer linear programme. Two dispatch policies distinguish unrestricted economic allocation from priority for current electricity sales. The preferred common configuration over six complete historical years contains electrolysis and no battery investment. Its value depends on hydrogen offtake and the right to divert electricity from direct sale. Contract sensitivity, annual sales floors, physical surplus attribution, chronological battery checks and independent numerical comparisons are used to assess this finding.

The Tier 2B specification couples multi-step forecasts of generation, demand and electricity prices to rolling-horizon optimisation with storage-state feedback. It separates forecast evaluation from decision evaluation and fixes investment before the test period. This draft contains the completed Tier 1 analysis and the Tier 2B method and evaluation specification; forecast and control experiment results remain to be added.

Contents

I Introduction and data 1
1 Introduction 1
2 Data audit and reconstruction 1
3 Common notation and assumptions 2

II Tier 1: deterministic investment appraisal 3
1 Problem analysis 3
2 Assumptions and research 3
3 Mathematical model 4
4 Results and analysis 5

III Tier 2B: forecasting and chronological operation 7
1 Information and experimental design 7
2 Multi-step forecasting model 7
3 Rolling optimisation and storage feedback 7
4 Evaluation and results to be completed 7

References R1

I. Introduction and data

1 Introduction

The project concerns a client's existing generation assets and the investment and operating decisions that can increase their value [1]. We study batteries and electrolysers in Tier 1 and select Tier 2B to examine chronological operation when future conditions must be predicted.

Our initial audit identified synchronised zero-generation records, missing dates, midnight snapshots used as daily means, abnormally low demand records and zero-volume price placeholders. The repair methods are full-day upstream reaggregation, short-gap interpolation, calibrated cross-source reconstruction and flagged whole-day imputation. These methods are distinguished in the data audit rather than treated as a single class of replacement.

Tier 1 combines research on electricity trading, batteries, electrolysers and financial conversion with mixed-integer linear programming. We compare allocation among direct electricity sale, storage and hydrogen by economic value with a policy that first sells all electricity for which current demand exists. Gain means additional receipts less fixed operation and annualised capital and renewals, against no investment under the same rule. With assumed hydrogen net receipts (after non-electricity variable costs) of £7.80/kg in 2020 pounds, a buyer accepting at most 10 tonnes per day and a 5% real discount rate, a common 20 MW electrolyser without batteries gives a mean annual equivalent gain of £10.94m under free allocation over 2020–25. Contract thresholds, alternative dispatch rights, individual years and battery chronology are then examined.

Tier 2B adds forecasts of generation, demand and prices to a rolling-horizon operating model. It repeatedly plans several days ahead, executes only the current day's decision and transfers the resulting battery stock to the next day. Figure I.1 shows the data, investment, prediction and state-feedback connections. Forecast accuracy and realised operating value will be evaluated separately; the experimental protocol is specified in Chapter III.

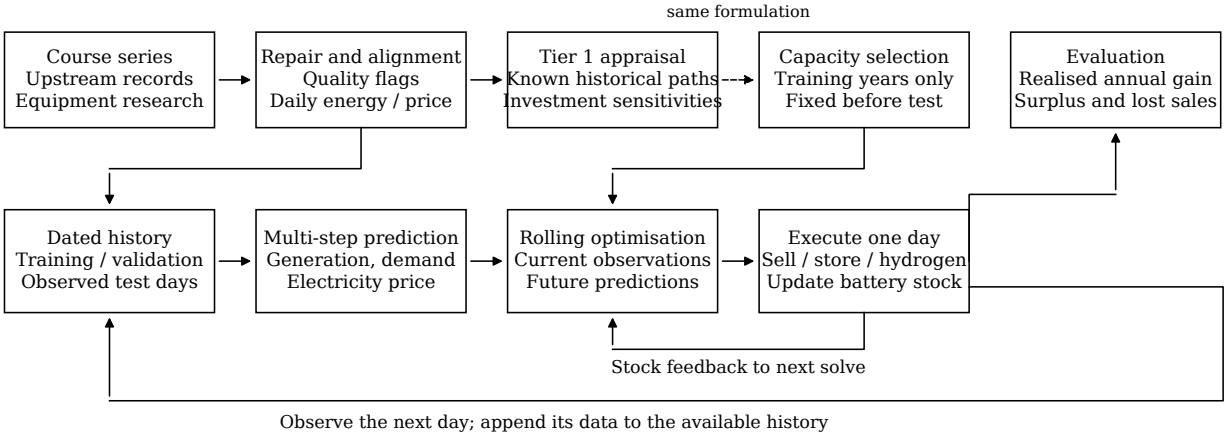


Figure I.1. Study architecture. Tier 1 uses known historical paths. Tier 2B uses past observations for forecasting and carries physical storage state forward; its investment decision precedes the test period.

2 Data audit and reconstruction

The five supplied files contain fuel-level generation, its total, simulated wind-farm generation, demand and market index prices. The fuel total is checked against its components; it is not added to them again. The separate simulated farm is added to that total as the declared asset boundary. Generation and demand are daily mean powers in MW, converted to MWh by multiplying by 24. We retain 1 January 2020–17 September 2026: 2,452 days in Coordinated Universal Time (UTC). Source checks use Elexon records and National Energy System Operator (NESO) series.

Table I.1. Detected problems, evidence and repair. Date counts can overlap.

Problem and extent	Identification and reconstruction
Eight non-wind/solar fuels zero on 107 days; solar zero on 42 days	Synchronous zeros conflict with archived records. Reaggregate Elexon actual generation by production type; use calibrated fuel half-hour or NESO solar series when that source is incomplete. Interpolate short gaps.
Midnight query endpoints: 7 generation, 350 demand, 346 price dates	Original values reproduce the first timestamp rather than the full day. Retrieve the complete UTC day and recalculate its mean.
Missing dates: 30 generation, 1 demand, 30 price	Calendar gaps are verified against the dated upstream series. Restore those records where available; otherwise use flagged whole-day estimates.
Invalid demand slots	Some five-minute records are below 10,000 MW, including 636 MW on 2 August 2022. Retain valid slots and reconstruct others from calibrated NESO demand; flag whole-day estimates.
Zero-volume price placeholders	115,571 provider records have no traded volume. Exclude those placeholders, volume-weight active providers within each half-hour, then average the day. Retain genuine zero and negative prices.

The repaired files share a complete calendar; fuel components sum to total generation, and the two price-unit columns retain their factor of 24. The 2 August demand mean is reconstructed as 28,744 MW. The audit identifies 436 dates with interpolation or estimation, 323 with cross-source reconstruction and 1,929 satisfying the strict quality flag; categories overlap [3]. The full set supports the primary analysis; the strict subset is a sensitivity check. All original values, replacements, source records and quality flags are retained. Historical prices are converted to 2020 pounds using monthly consumer price indices [4].

3 Common notation and assumptions

Only quantities shared by both tiers are defined here. Equipment decisions, dispatch policies, forecast information and technical parameters are introduced in their respective chapters.

Table I.2. Notation shared across the study.

Symbol	Meaning	Unit
$t \in \mathcal{T}$	UTC day in the selected study period	day
G_t, D_t	Available generation and market demand (sales ceiling)	MWh
p_t	Market electricity price, in constant 2020 pounds	£/MWh
S_t	Physical surplus, $\max(G_t - D_t, 0)$	MWh
Y	Represented years: sum of each sampled day's calendar-year fraction	year

The supplied series describe a constructed system rather than observed network congestion [1, 2]. Existing generation is fixed; its unchanged costs cancel in investment comparisons. Demand limits sales and is not an additional unpaid internal load. We assume no electricity imports, no additional export outlet and no payment for unused generation. Spatial transfer limits are omitted. All monetary comparisons use constant 2020 pounds. These conventions are common to both tiers.

II. Tier 1: deterministic investment appraisal

1 Problem analysis

Given the paths in Chapter I, we choose equipment capacity and daily use. Electricity can be sold, stored, converted to hydrogen or left unused. The objective is equivalent annual profit gain relative to no investment under the same operating rules, after capital, fixed operation and renewal costs.

Open choices concern displacement of electricity sales, hydrogen offtake and value, equipment cost scope, discount rate and battery chronology. We specify these below and test the choices that change the investment.

2 Assumptions and research

2.1 Tier-specific assumptions

Investment is once at the start, in whole reference modules. Electrolysers can be off or run at 20–100% of online module power. Hydrogen is sold directly, with no storage or reconversion. Its netback (receipts after non-electricity variable costs) is assumed to equal a rebased support value. The 10 t/day buyer limit and 5% real discount rate are analytical assumptions.

We distinguish *policy A*, which allocates generation freely by economic value, and *policy B*, which first sells $\min(G_t, D_t)$ each day. B sells even at negative prices. Each has its own zero-investment comparator. Battery energy is balanced over the sample, with at most one usable window per day; fixed nonzero batteries are also checked chronologically. This follows the simplified Tier 1 time treatment [1, 2].

2.2 Research objectives and parameter selection

Research establishes market boundaries, equipment scale and performance, cost scope and financial comparability (Table II.1). The battery module is a research reference; the brief specifies its two-hour duration.

Table II.1. Technical and financial research used in Tier 1. Costs are in 2020 pounds.

Research objective	Selected input	Source and evidence scope
Battery performance and scale	1.5 MW / 3 MWh module; 20–80% state of charge; 90% round-trip efficiency; 15-year life	Brief and Danish Energy Agency technology catalogue; reference equipment, not a supplier offer [1, 5].
Battery cost	£256,235/MWh capital; £14,862/MW-year fixed operation	Brief's EUR 288,000/MWh converted with 2020 exchange rate; 2.9% annual maintenance convention [9].
Electrolysis operation	20 MW modules; 20–100% online load; 55.9 kWh/kg; 30-year life	Brief technical reference; life and renewal assumptions from UK production cost estimates [1, 7].

Table II.1. Technical and financial research (continued).

Research objective	Selected input	Source and evidence scope
Electrolyser costs	£29.2m per module; £0.517m/year fixed cost; £9.792m stack renewal in years 11 and 22	Installed-cost survey midpoint; separate absolute UK fixed and stack proxies, rather than applying an old stack percentage to new capital [6, 7].
Hydrogen outlet and value	Direct sale; £7.80/kg netback; 10,000 kg/day	UK hydrogen support reference rebased to 2020; realised-netback and buyer-cap scenarios, not a verified market contract [8, 9].
Financial comparability	Monthly inflation adjustment; 5% real discount; zero terminal value	Official inflation and exchange-rate series; discount-rate sensitivity and assumed zero residual value [4, 9].

The brief's £9.50/kg hydrogen benchmark has no explicit price base. We convert the published £175/MWh in 2012 pounds using 39.4 kWh/kg and the 2020/2012 consumer-price-index ratio. The resulting netback assumes no further non-power deduction. Electrolyser capital uses the midpoint of the published US$2,000–2,600/kW installed range. These reference estimates motivate the hydrogen-value, offtake, capital and discount sensitivities.

3 Mathematical model

Table II.2. Tier 1 notation, additional to Chapter I.

Symbol	Meaning	Unit or domain
b, n	Installed battery and electrolyser modules	$\mathbb{Z}_{\geq 0}$
z_t	Online electrolyser modules on day t	$\mathbb{Z}_{\geq 0}$
q_t, c_t, r_t	Direct sale, battery charge and discharge	MWh
h_t, u_t	Electrolysis electricity input and unused generation	MWh
v, L, κ	Hydrogen netback, daily buyer cap and electricity per kg	£/kg; kg/day; MWh/kg
E_B, η	Usable battery module energy and one-way efficiency	MWh; dimensionless
$E_H^{\min}, E_H^{\max}$	Online module's daily input limits	MWh
a_B, a_H	Equivalent annual costs per installed module	£/year

Here $\kappa = 0.0559$, $E_B = 1.8$, $\eta = \sqrt{0.90}$ and $(E_H^{\min}, E_H^{\max}) = (96, 480)$, in the units above. Let x collect the installed capacities and daily decisions. The annual profit before subtracting the constant no-investment comparator is

$$\max_x \ \Pi(x) = \frac{1}{Y} \sum_{t \in \mathcal{T}} \left[p_t(q_t + r_t) + \frac{v h_t}{\kappa} \right] - a_B b - a_H n. \tag{II.1}$$

The feasible set is defined by

$$q_t + c_t + h_t + u_t = G_t, \qquad q_t + r_t \leq D_t, \tag{II.2a}$$

$$c_t \leq E_B b/\eta, \qquad r_t \leq \eta E_B b, \tag{II.2b}$$

$$\sum_t r_t = \eta^2 \sum_t c_t, \tag{II.2c}$$

$$E_H^{\min} z_t \leq h_t \leq E_H^{\max} z_t, \qquad h_t/\kappa \leq L, \qquad z_t \leq n, \tag{II.2d}$$

$$q_t, c_t, r_t, h_t, u_t \geq 0, \qquad b, n, z_t \in \mathbb{Z}_{\geq 0}. \tag{II.2e}$$

All day-indexed constraints hold for every $t \in \mathcal{T}$. Equation (II.2a) allocates generation and limits total electricity sales. Equations (II.2b)–(II.2c) link battery throughput to installed capacity and impose the round-trip loss. Equation (II.2d) enforces online loading, daily hydrogen acceptance and installed capacity; zero online modules allow shutdown. Equation (II.2e) gives nonnegativity and integrality. Policy B adds $q_t = \min(G_t, D_t)$.

For asset $j \in \{B, H\}$, equivalent annual cost is

$$a_j = F_j + \frac{i}{1-(1+i)^{-N_j}} \left(K_j + \sum_{k \in \mathcal{R}_j} \frac{C_{jk}}{(1+i)^k} \right), \tag{II.3}$$

where F_j is fixed annual operation, K_j initial capital including commissioning, N_j life, i discount rate and C_{jk} renewal cost in the years $\mathcal{R}_j$. At $i = 0$, the capital-recovery factor is $1/N_j$. The reported gain is $\Pi(x) - \Pi(x_0)$, where x_0 is the same-policy zero-investment solution. The implementation subtracts this constant during optimisation so that background electricity receipts do not mask small investment differences.

4 Results and analysis

4.1 Capacity and historical performance

We optimise one common equipment size over the six complete years 2020–25, then replay that fixed size in each year. Both policies select one 20 MW electrolyser and no batteries. Table II.3 reports all complete years. The 2026 partial year is archived separately. Among complete years, 2023 has the fewest surplus days and least physical surplus, so it is used for the detailed stress-year comparisons below.

Table II.3. Fixed 20 MW electrolysis across complete years. Surplus and capture are GWh/year; gains are £m/year relative to each policy's zero investment. Capture is identical in A and B.

Year	Surplus days	Surplus	Capture	A gain	B gain
2020	92	2067	43.9	16.78	3.12
2021	43	911	20.6	6.29	−0.13
2022	154	4240	73.9	8.91	7.31
2023	29	550	13.9	8.57	−1.07
2024	30	1229	14.4	11.95	−1.00
2025	112	4660	53.7	13.17	4.48
Mean	–	2276	36.7	10.94	2.12

The positive mean under sales priority includes three losing years. For 2023 alone, battery-only optimisation selects zero capacity in both policies; hydrogen-only and joint choices coincide. B's freely chosen investment is zero, while forcing the same 20 MW asset gives the loss in Table II.3. A sale-first rule on that asset reproduces B, providing a capacity-matched simple strategy comparison. A's improvement therefore comes from diverting saleable power rather than installing different equipment.

4.2 Energy attribution and contract sensitivity

In 2023 the electrolyser takes 171.8 GWh: 13.92 GWh from physical surplus and 157.9 GWh diverted from direct sale. Thus only 8.1% of its input is surplus capture, equivalent to 2.53% of that year's 549.9 GWh surplus. The 29 surplus days each use the full 0.48 GWh module input. The 10 t/day buyer cap can accept at most 0.559 GWh/day of electrolysis input, limiting pure surplus capture to 16.21 GWh even with larger equipment.

Fuel aggregates do not identify which generator supplied electrolysis. Under surplus-first attribution, define captured energy $C_t = \min(h_t, S_t)$ and wind-plus-solar generation R_t. Its contribution W_t to capture satisfies

$$\max\{0, C_t - (G_t - R_t)\} \le W_t \le \min\{C_t, R_t\}. \tag{II.4}$$

The annual interval is 0–13.92 GWh; adding biomass and run-of-river hydro to the renewable category leaves these bounds unchanged.

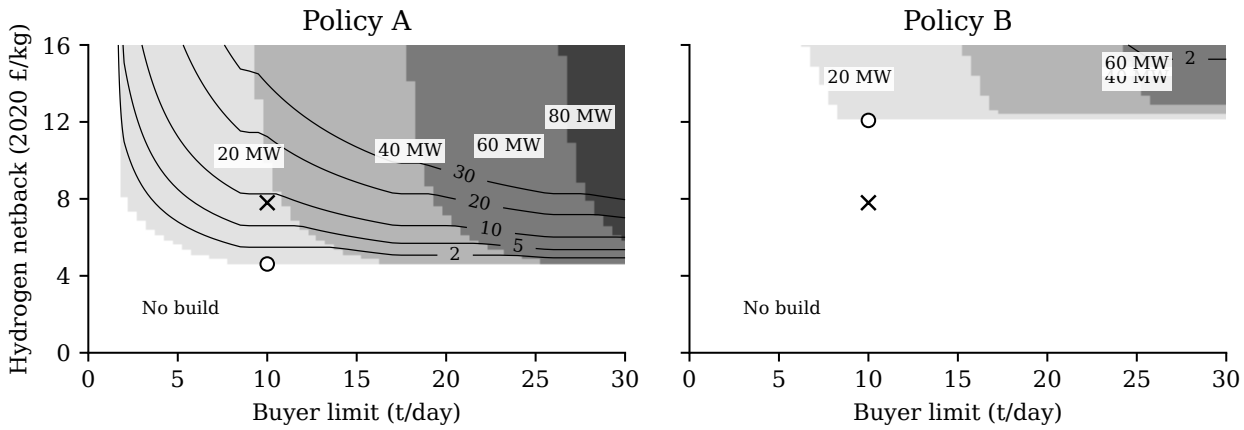


Figure II.1. 2023 optimal capacity over 7,930 contract combinations. Contours are annual gain (£m); the cross marks the reference contract and the circle the fixed 20 MW break-even at 10 t/day.

At 10 t/day, the 20 MW break-even netback is £4.61/kg in A and £12.08/kg in B. At £7.80/kg, A first breaks even at 2.23 t/day. With the reference cap, 40 MW lowers A's gain to £7.47m/year; at 50 t/day the selected capacity rises to 120 MW, with £49.09m/year gain and £175.3m capital. The buyer's daily acceptance therefore affects scale as well as profitability.

Table II.4. Fixed 20 MW comparisons, policy A. Sales lost are realised GWh/year; gains are £m/year. All use 2023 except the stated 2025 downside.

Comparison	Sales lost	Gain
Restore half of A's lost sales	79.0	5.19
Retain 99.99% of annual sales	21.5	1.04
Capital +25%, discount 8%	157.9	7.29
2025: £4/kg, 5 t/day buyer	49.2	−0.24

Annual sales floors let the operator preserve selected electricity sales; the 99.99% floor binds, unlike a general statement that sales have priority. The cost stress holds stack-renewal cost fixed. The strict-quality 2023 subset keeps the original investment choices.

4.3 Validation and conclusion

The six-year battery revenue bound, even with free charging, is £54.0k per module-year in A, below the £96.4k annual cost. This rules out batteries in the common-size model. For one fixed battery in 2023 A, replacing pooled energy with daily stock reduces operating value from £13.3k to £4.3k. The stock check uses usable stock $s_t = s_{t-1} + \eta c_t - r_t/\eta$, $0 \le s_t \le E_B b$, with zero initial and terminal stock.

All 60 exported Xpress schedules reconcile flow and cash, with physical residuals below 0.001 MWh; eight contract points match exact enumeration. All 14 annual replays agree with an independent implementation to its published precision [10]. The 2023 A gain decomposes into £23.97m hydrogen receipts, less £12.39m forgone electricity, £0.52m fixed operation and £2.49m annual capital and stack cost. Repeating that year's operating cash for 30 years gives a net present value of £131.8m, with simple/discounted paybacks of 2.64/2.90 years, including stack replacements and zero residual value.

The evidence supports assessing a 20 MW electrolyser against actual offtake and dispatch terms before committing £29.2m. Sales priority must retain its loss-year risk in that assessment. The battery conclusion applies to the specified daily energy-shifting service and costs; intraday and ancillary-market revenues are outside this model. Chapter III therefore examines information and chronology separately rather than treating the Tier 1 optimum as a forecast-based operating policy.

III. Tier 2B: forecasting and chronological operation

1 Information and experimental design

The operating decision is made once per day. Following the classroom simplification, current-day generation, demand and price are known at that decision, while future days are unknown [2]. Let $y_t = (G_t, D_t, p_t)$ and let $\mathcal{I}_t$ contain observations through day t, the installed equipment and the previous day's usable stock. Only information in $\mathcal{I}_t$ can determine the implemented action. Current-day aggregates are treated as observed at this daily planning decision.

We reserve 2020–22 for training and capacity selection, 2023 for model and horizon selection, and 2024–25 for the final test. A forecasting training target must lie before its corresponding cutoff. Repair calibration, scaling and hyperparameter selection use training/validation data only. The six-year Tier 1 capacity comparison remains a hindsight benchmark; test observations must not choose the capacity used in the predictive experiment.

2 Multi-step forecasting model

The proposed first predictive model is regularised linear autoregression, compared against a seven-day seasonal-naive predictor. At each forecast origin t, the feature vector f_t contains observed generation, demand and price lags, day-of-week indicators and annual sine/cosine terms. For variable $j \in \{G, D, p\}$ and horizon $k = 1, \ldots, H$, fit

$$\widehat{\beta}_{jk} = \arg\min_{\beta} \sum_{\tau \in \mathcal{A}_k} (y_{\tau+k,j} - f_\tau^\top \beta)^2 + \lambda \|\beta\|_2^2, \qquad \widehat{y}_{t+k,j} = f_t^\top \widehat{\beta}_{jk}, \tag{III.1}$$

where $\mathcal{A}_k$ contains eligible training origins whose targets are already observed, $\lambda \geq 0$ is the validation-selected penalty and H is the planning horizon. Forecast generation and demand are projected onto nonnegative values; negative price forecasts remain allowed. The seasonal-naive forecast uses the most recent observation with the same weekday, recursively if needed.

3 Rolling optimisation and storage feedback

At day t, installed battery and electrolyser counts are fixed. A planning problem uses observed values for the current day and predictions for $t+1, \ldots, t+H$. It retains the Tier 1 flow, demand, throughput and electrolysis constraints, but replaces the pooled energy balance with chronological stock:

$$s_\tau = s_{\tau-1} + \eta c_\tau - r_\tau/\eta, \qquad 0 \leq s_\tau \leq E_B b, \qquad \tau = t, \ldots, t+H. \tag{III.2}$$

Here s_{t-1} is the measured initial stock for that solve. The optimiser maximises forecast operating receipts plus a terminal-stock value; the latter is selected on validation data to avoid repeatedly liquidating storage at an artificial short horizon. Only (q_t, c_t, r_t, h_t) is implemented. Its realised stock becomes the next solve's initial condition. The test's initial and final stock treatment is held equal across all compared policies.

4 Evaluation and results to be completed

Forecast error is reported by series and horizon using mean absolute error and root mean square error, with price errors in £/MWh and physical errors in MWh. Operating evaluation compares the predictive controller, the same controller with seasonal-naive forecasts, a current-day rule and a full-path perfect-information optimiser, all at identical installed capacities and with identical data, costs and terminal stock. The gap to perfect information measures the realised loss from restricted information, rather than a change in investment scale.

Since zero batteries remove cross-day storage, fixed nonzero battery cases will diagnose prediction and feedback value alongside the training-selected investment. Results will include forecast errors, realised annual value, surplus capture, lost sales, stock feasibility and seasonal/adverse-period comparisons. These forecast and control experiments are pending.

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